



IT3051 – Fundamentals of Data Mining

Mini Project Assignment 2026

Dataset Proposal

Air Quality Risk Classification Using EPA Historical Monitoring Data

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1. Problem Scenario

1.1 Background and Motivation

Air pollution is a major public health concern, contributing to respiratory and cardiovascular illness, and disproportionately affecting vulnerable groups such as children, the elderly, and people with pre-existing conditions. In the United States, the Environmental Protection Agency (EPA) has operated a nationwide network of ambient air-monitoring stations for decades, tracking pollutants such as fine particulate matter (PM_{2.5}), ozone, carbon monoxide, nitrogen dioxide, and sulfur dioxide against National Ambient Air Quality Standards (NAAQS). This monitoring network produces a long, consistent historical record that health agencies use to understand pollution trends, issue public advisories, and evaluate the impact of pollution control policy.

Raw pollutant measurements are not always easy for non-specialist users to interpret. The EPA Air Quality Index (AQI) translates concentrations of criteria pollutants, including PM_{2.5}, into common health-related categories such as Good, Moderate, Unhealthy for Sensitive Groups, Unhealthy, Very Unhealthy, and Hazardous. EPA's current PM_{2.5} AQI reference table specifies concentration breakpoints for these categories and the corresponding AQI ranges. A data-mining system that predicts the following year's PM_{2.5} risk level from historical site information could provide an analytical early-warning indicator for environmental monitoring and planning. The system would not replace official EPA or public-health advisories; instead, it would demonstrate how historical environmental data can be transformed into a supervised machine-learning problem.

1.2 Problem Statement

Given a monitoring site's historical annual PM_{2.5} statistics and contextual site information, there is a need for a predictive system that classifies the likely PM_{2.5} risk level for the following year at that site. The objective is to use only information available before the prediction year, so that the evaluation represents a genuine forecasting-style task rather than a model that indirectly observes the future target.

1.3 Prediction Objective

Primary objective:

Build a supervised classification model that predicts a monitoring site's next-year PM_{2.5} air-quality health risk category, based on that site's own historical pollutant statistics and contextual features.

Secondary objectives:

- Provide a confidence score alongside the predicted category so that public health agencies can prioritise sites with the highest-confidence elevated-risk predictions.
- Identify which historical indicators (e.g., prior-year concentration trend, exceedance history, seasonal patterns) most strongly predict a worsening risk category.
- Compare multiple algorithms and preprocessing strategies systematically to justify a final model choice with evidence, per Stages 6–7 of the assignment.
- Present the prediction through a simple interface usable by a local public health officer, not an air-quality scientist.

1.4 Target Variable

Attribute	Value	Meaning
Name	pm25_risk_category_next_year (engineered)	Categorical risk level derived from the following year's PM2.5 concentration using a documented EPA PM2.5 AQI breakpoint mapping.
Type	Multi-class (categorical)	Expected categories: Good, Moderate, Unhealthy for Sensitive Groups, Unhealthy, Very Unhealthy, and potentially Hazardous, subject to observed class frequencies.
Business meaning	Early-warning value	A higher-risk prediction can identify monitoring sites whose historical pattern warrants closer attention.

Important target-definition note: the original Kaggle dataset does not contain pm25_risk_level_next_year. It is an engineered target. Because EPA AQI is based on specific concentration averaging periods, the project will document the exact PM2.5 concentration field and breakpoint mapping used rather than treating the annual arithmetic mean as a direct AQI value. EPA's current AQI table defines PM2.5 breakpoints using 24hour concentration ranges.

1.5 Task Type and Justification

Type: Supervised multi-class classification (with the option to simplify to a binary “elevated risk vs. not” framing if class sizes for the most severe categories prove too small.)

Justification: The proposed output is a categorical risk level rather than an exact continuous pollutant value. Classification therefore matches the intended decision-support output. The target will be constructed from a future year's PM2.5 concentration, while model features will be restricted to the prediction year and earlier. Using future-year predictor measurements would constitute data leakage and will be explicitly avoided.

1.6 Key Stakeholders

Stakeholder	Role	Decisions Supported
Public Health Departments	Issue public air-quality advisories	Decide when and where to issue early health warnings for vulnerable groups
Local Environmental Agencies	Monitor and regulate local pollution sources	Prioritize inspection or enforcement in areas trending toward higher risk

Schools and Care Facilities	Protect vulnerable populations (children, elderly)	Decide whether to limit outdoor activity on anticipated high-risk days
Urban Planning / Policy Makers	Plan long-term pollution-control policy	Evaluate whether current interventions are improving projected risk trends
General Public (via advisory app)	Everyday health and activity planning	Decide on outdoor activity levels based on anticipated air-quality risk

1.7 User Requirements

Functional Requirements

- Accept a monitoring site's historical annual pollutant statistics and return a predicted next-year risk category with a confidence score.
- Support batch prediction across multiple monitoring sites for a regional overview.
- Allow a configurable sensitivity setting, since a school or care facility may want to be alerted at a lower risk threshold than a general public-facing app.
- Present a short, plain-language explanation of the main factors behind a given prediction (e.g., rising 3-year concentration trend, prior-year exceedance).

Non-Functional Requirements

- Reasonable response time appropriate for an annual/seasonal planning tool rather than a millisecond-level real-time system.
- Usability for non-specialist public health and planning staff, avoiding unnecessary atmospheric-science terminology.
- Robustness to sites with incomplete monitoring history (not every site has data for every year).
- Reproducibility: the exact BigQuery extraction query, pollutant scope (PM2.5), and date range used to build the training dataset must be documented so it can be re-run as new annual data becomes available.

1.8 Expected Inputs and Outputs

Input (per monitoring site, per year):

- Current and prior years' PM2.5 annual summary statistics, such as arithmetic mean, standard deviation, maximum and percentile measures.
- Observation count/completeness measures and applicable exceedance-related statistics.
- Site metadata such as state, county, latitude and longitude.
- Engineered historical features such as year-over-year change and rolling multi-year averages, calculated only from past information.

Output:

- A predicted risk category for the following year (e.g., Good, Moderate, Unhealthy for Sensitive Groups, Unhealthy).
- A confidence score for the predicted category.
- A short, plain-language recommendation appropriate to the predicted category.

Example output (illustrative):

Field	Example Value
site	County Monitoring Site #0021
prediction	Unhealthy for Sensitive Groups
confidence	0.68
recommended_action	Advise sensitive groups to limit prolonged outdoor exertion next year's high-pollution season

2. Main Dataset Proposal

2.1 Dataset Name and Original Source

Air Quality Annual Summary (Kaggle dataset “epa/air-quality”), sourced from the U.S. Environmental Protection Agency's (EPA) Air Quality System (AQS), and made available as a Google BigQuery public dataset accessible through Kaggle. It summarises air-quality monitoring data collected from EPA and state/local monitoring stations across the United States.

2.2 Dataset URL, Access Method, and Citation

Kaggle dataset page: <https://www.kaggle.com/datasets/epa/air-quality>

Underlying data source: US Environmental Protection Agency, Air Quality System (AQS) Data Mart, <https://www.epa.gov/outdoor-air-quality-data>

Citation: U.S. Environmental Protection Agency. Air Quality System / AirData annual air-quality summary data. Dataset accessed through Kaggle: “Air Quality Annual Summary.”

Access method: The Kaggle dataset provides a large annual-summary data file. The group will create the modelling dataset by selecting the PM2.5 records and relevant fields, then resolving any multiple-record siteyear cases according to documented preprocessing rules. EPA also provides annual summary files and APIs; EPA notes that annual summary data are at monitor level and may contain more than one entry per monitor because of differences such as sample duration and pollutant standards.

Licence: The Kaggle dataset is listed as CC0: Public Domain. The underlying source is U.S. government environmental monitoring data.

2.3 Description and Real-World Context

The dataset contains historical annual air-quality information covering 1987–2017. The original dataset includes 55 columns describing monitoring sites, pollutants, annual measurements, maxima, percentile statistics, observation counts, completeness measures, exceedance information, and related metadata. The dataset therefore provides a credible real-world environmental foundation for studying how historical PM2.5 patterns can be used in a supervised classification problem.

2.4 Proposed Dataset Construction Plan

Pollutant Scope	PM2.5 (fine particulate matter) — chosen as the single pollutant most consistently linked to a standard, welldocumented AQI breakpoint table and to public health outcomes
Geographic Scope	All available US monitoring sites, or a defined subset of states, to be confirmed once the extracted row count is known
Time Range	All available years of PM2.5 monitoring within the dataset's 1987–2017 coverage (PM2.5 monitoring became widespread from around the late 1990s onward, so the effective usable range is expected to be narrower and will be confirmed during extraction)
Aggregation Level	One row per monitoring site per year (as provided by the annual summary table)
Extraction Method	A single SQL query filtering the annual_summary table to parameter_name = 'PM2.5', selecting the relevant statistical and metadata columns, run via Kaggle's BigQuery integration and exported to CSV

2.5 Candidate Features

Feature(s)	Type	Description
state_name / county_name	Categorical	Geographic identifiers for the monitoring site
latitude / longitude	Numeric (continuous)	Precise site location, usable for spatial feature engineering
year	Numeric (integer)	Calendar year of the summary record
arithmetic_mean	Numeric (continuous)	Annual mean PM2.5 concentration at the site; used to engineer the risk-category target for the following year, and as a historical feature for earlier years
arithmetic_standard_dev	Numeric (continuous)	Variability of PM2.5 readings within the year
first_max_value / 90th_percentile / 98th_percentile	Numeric (continuous)	Peak and high-percentile concentration statistics, useful for capturing short-term pollution spikes

primary_exceedance_count	Numeric (integer)	Number of days the site exceeded the primary NAAQS standard that year — a
		strong historical risk indicator when used from prior years only
observation_percent	Numeric (continuous)	Data-completeness indicator for the year, useful for filtering out unreliable lowcoverage records
pm25_risk_category_next_year (target, engineered)	Categorical	AQI health-risk category derived from next year's arithmetic_mean using EPA's standard PM2.5 breakpoint table

2.6 Expected Machine Learning Task and Modelling Plan

Task: Multi-class classification (next-year PM2.5 risk category), with a simplified binary variant (elevated risk vs. not) as a fallback if some categories prove too rare after extraction.

Candidate Algorithms

Algorithm	Rationale
Logistic Regression (multinomial)	Interpretable baseline showing the directional effect of trend and exceedancehistory features on next-year risk.
Random Forest	Captures non-linear relationships between historical concentration trends and future risk category, robust to mixed feature scales.
Gradient Boosting (XGBoost/LightGBM)	Typically strong on structured, moderately imbalanced tabular data; supports class weighting for rarer high-risk categories.
Support Vector Machine (SVM)	Margin-based comparison point after scaling continuous concentration and trend features.
k-Nearest Neighbours — optional 5th model	Included as a simple, distance-based comparison, given nearby sites and years may have genuinely similar pollution profiles.

Validation Strategy

- Chronological (time-based) train/test split by year, rather than a random split, so the model is evaluated on genuinely future years it has not seen, avoiding the same kind of look-ahead leakage discussed in the group's earlier rainfall-forecasting proposal.
- Grouped validation by monitoring site where appropriate, to avoid the same physical site appearing in both training and test folds in a way that leaks site-specific baseline levels.
- All feature engineering (e.g., trend calculations, rolling averages) computed using only past information relative to each prediction year, to avoid look-ahead leakage.

Evaluation Metrics

Metric	Rationale
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Macro-averaged F1score	Treats every risk category equally regardless of size, important given higherrisk categories are expected to be rarer than ‘Good’ or ‘Moderate’.
Recall on higher-risk categories	Prioritized specifically, since missing a genuinely elevated-risk year has greater public health cost than a false alarm.
Confusion Matrix	Shows which adjacent risk categories (e.g., Moderate vs. Unhealthy for Sensitive Groups) are most often confused, directly supporting the report's interpretation section.
Overall Accuracy (secondary)	Reported alongside the above but not relied upon alone, since class sizes are expected to be uneven across risk categories.

2.7 Suitability for the Assigned Problem

- Directly relevant, socially meaningful scenario: forecasting air-quality risk is a genuine public-health and environmental-policy use case with clear stakeholders.
- Sourced from a highly credible, authoritative source (the US EPA's official air-monitoring system), strengthening the dataset's real-world credibility for the technical report.
- Multi-decade historical coverage (1987–2017) supports genuine trend-based feature engineering (multi-year rolling averages, year-over-year change) that a single-snapshot dataset could not offer.
- Being self-extracted via a defined SQL query gives the group full control over pollutant scope, geographic scope, and feature set, allowing the dataset to be tailored precisely to a leakage-safe, forecasting-style scenario.
- Site-level and year-level granularity supports meaningful EDA around geographic and temporal patterns in air pollution.

2.8 Data-Quality Observations and Limitations

- The dataset's coverage ends in 2017, so the model would be forecasting based on a historical period and would not reflect any changes in pollution patterns or regulation since then; this is an explicit limitation to state in the report.
- Because access requires a SQL extraction step rather than a ready-made file, the group is responsible for documenting the exact query used and verifying the resulting row count and class distribution during EDA, rather than assuming figures from secondary sources.
- Not every monitoring site has a complete, unbroken annual history; sites with sparse or inconsistent reporting will need to be filtered or handled explicitly, and this decision should be justified in the preprocessing write-up.
- The rarest risk categories (e.g., ‘Very Unhealthy’ or ‘Hazardous’) may have very few site-years in the extracted data, which may require merging adjacent categories or focusing evaluation on a simplified elevated-risk-vs-not framing, similar to the small-class handling discussed for the group's RT-IoT2022 proposal.
- A chronological, not random, train/test split is essential, and grouping by site should also be considered, since a naive random split could let the model see future information from the same physical site.

2.9 Ethical, Privacy, and Licensing Considerations

- No personal or individually identifiable data is involved; the dataset consists entirely of environmental monitoring statistics at fixed public monitoring sites.

- EPA data is US government public-domain information, freely available for reuse without licensing restrictions.
- As with any public-health forecasting tool, the group will note in its limitations section that predictions are intended to support, not replace, official EPA and local public health agency air-quality advisories in any real deployment.